Chapter 10. Applications of Theta Functions

Jacobi triple product, transformation laws, partitions, the two- and four-squares theorems

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Overview

The *theta function*

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

converges for all $z \in \mathbb{C}$, $\tau \in \mathbb{H}$. In z it is elliptic-like (period 1, "quasi-period" τ); in τ it is modular, connected with partitions and sums of squares. The tools are the *triple product formula* and the *transformation law*; applications are partitions and the two- and four-squares theorems.

1 Product formula for the Jacobi theta function

Motivation. $\Theta(z|\tau)$ is entire with explicit zeros, and its product formula yields partition and sums-of-squares identities.

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, **Jacobi's theta function** is

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}. \tag{1}$$

Special cases: $\theta(\tau) = \Theta(0|\tau) = \sum_n e^{\pi i n^2 \tau}$, $\vartheta(t) = \theta(it)$; the heat kernel on the circle (Book I) is $H_t(x) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = \Theta(x|4\pi i t)$; ϑ's functional equation gave ζ's (Chapter 6).

Proposition 1.1

The function Θ satisfies the following properties:
(i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
(ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
(iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$.
(iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. If $\mathrm{Im}(\tau) = t \geq t_0 > 0$ and $|z| \leq M$, then $\sum_n |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty$, uniformly; this gives (i).

(ii) is clear since $e^{2\pi i n z}$ has period 1. For (iii), complete the square:

$$\Theta(z+\tau|\tau) = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} = \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1) z} e^{-2\pi i z} = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}.$$

By (ii) and (iii) it suffices for (iv) to show $\Theta(1/2+\tau/2|\tau) = 0$: $\Theta(1/2+\tau/2|\tau) = \sum_n (-1)^n e^{\pi i (n^2+n)\tau}$, and pairing $n \geq 0$ with $-n-1$, which has opposite parity and $(-n-1)^2 + (-n-1) = n^2 + n$, the terms cancel. ∎

With the standard notation $q = e^{\pi i \tau}$, the **triple product** is

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 1.2

The function $\Pi(z|\tau)$ satisfies the following properties:
(i) $\Pi(z|\tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
(ii) $\Pi(z+1|\tau) = \Pi(z|\tau)$.

(iii) $\Pi(z+\tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
(iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\mathrm{Im}(\tau) \geq t_0 > 0$, $|q| \leq e^{-\pi t_0} < 1$ and each factor is $1+O(|q|^{2n-1}e^{2\pi|z|})$; since $\sum|q|^{2n-1} < \infty$, the results on infinite products (Section 3 of Chapter 5) give (i).

(ii) is clear. For (iii), since $q^2 = e^{2\pi i\tau}$,

$$\Pi(z+\tau|\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n+1}e^{2\pi iz})(1+q^{2n-3}e^{-2\pi iz}),$$

and comparing with $\Pi(z|\tau)$, $\Pi(z+\tau|\tau) = \Pi(z|\tau)\frac{1+q^{-1}e^{-2\pi iz}}{1+qe^{2\pi iz}}$;

(iii) follows from $(1+x)/(1+x^{-1}) = x$. For (iv), a convergent product vanishes only where a factor does, and $1-q^{2n} \neq 0$. $1+q^{2n-1}e^{2\pi iz} = 0$ iff $(2n-1)\tau + 2z \equiv 1 \pmod 2$, i.e. $z \equiv 1/2+\tau/2-n\tau \pmod 1$; $1+q^{2n-1}e^{-2\pi iz} = 0$ iff $z \equiv 1/2+\tau/2+(n-1)\tau \pmod 1$ $(n \geq 1)$. These are all the zeros, simple as $e^w - 1$ has a simple zero at 0. ∎

Theorem 1.3 (Product formula)

For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.

Proof. Fix $\tau \in \mathbb{H}$. There is a constant $c(\tau)$ with

$$\Theta(z|\tau) = c(\tau)\Pi(z|\tau)\,: \tag{2}$$

$F(z) = \Theta(z|\tau)/\Pi(z|\tau)$ is entire (by (iv) of both propositions) and doubly periodic with periods 1, τ (by (ii), (iii)), hence constant. We show $c(\tau) = c(4\tau)$. Put $z = 1/2$ in (2) ($e^{\pm 2\pi iz} = -1$):

$$\sum_n(-1)^n q^{n^2} = c(\tau)\prod_{n=1}^{\infty}(1-q^{2n})(1-q^{2n-1})^2 = c(\tau)\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1}), \quad c(\tau) = \frac{\sum_n(-1)^n q^{n^2}}{\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1})}. \tag{3}$$

Put $z = 1/4$ ($e^{2\pi iz} = i$): $\Theta(1/4|\tau) = \sum_n q^{n^2}i^n$, and since $1/i = -i$, only even $n = 2m$ survive: $\Theta(1/4|\tau) = \sum_m q^{4m^2}(-1)^m$. On the other hand

$$\Pi(1/4|\tau) = \prod_{m=1}^{\infty}(1-q^{2m})(1+iq^{2m-1})(1-iq^{2m-1}) = \prod_{m=1}^{\infty}(1-q^{2m})(1+q^{4m-2}) = \prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4}),$$

the last step by splitting $2m = 4n$ and $2m = 4n-2$ in the first factor ($(1-q^{4n-2})(1+q^{4n-2}) = 1-q^{8n-4}$). Hence

$$c(\tau) = \frac{\sum_n(-1)^n q^{4n^2}}{\prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4})}, \tag{4}$$

and comparing (3) and (4) ($q \mapsto q^4$ corresponds to $\tau \mapsto 4\tau$), $c(\tau) = c(4\tau)$. So $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{\pi i 4^k\tau} \to 0$, (3) gives $c(4^k\tau) \to 1$; so $c(\tau) = 1$. ∎

Corollary 1.4

If $\mathrm{Im}(\tau) > 0$ and $q = e^{\pi i\tau}$, then

$$\theta(\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

Corollary 1.5

For each fixed $\tau \in \mathbb{H}$, the quotient

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ, and with a double pole at $z = 1/2+\tau/2$.

(Primes denote differentiation in z.)

Proof. Let $F = \Theta'/\Theta$. Differentiating (ii) and (iii) of Proposition 1.1 gives $F(z+1) = F(z)$, $F(z+\tau) = F(z) - 2\pi i$, so F' is doubly periodic; Θ has only a simple zero at $1/2 + \tau/2$ in the fundamental parallelogram, so F' has only a double pole. ∎

1.1 Further transformation laws

In Chapter 9 the modular character was expressed through $T_1 : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. For Θ it is natural to use $T_2 : \tau \mapsto \tau + 2$ and S, since $\Theta(z|\tau+2) = \Theta(z|\tau)$ but $\Theta(z|\tau+1) \neq \Theta(z|\tau)$.

Theorem 1.6

If $\tau \in \mathbb{H}$, then

$$\Theta(z|-1/\tau) = \sqrt{\tau/i}\, e^{\pi i z^2 \tau}\Theta(z\tau|\tau) \qquad \text{for all } z \in \mathbb{C}. \tag{5}$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to take $z = x$ real and $\tau = it$: both sides are holomorphic in τ for fixed real x, and entire in z for fixed τ. Then (5) reads

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a, this is $\sum_n e^{-\pi t(n+a)^2} = \sum_n t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}$, which is equation (3) of Chapter 4 (from Poisson summation). ∎

Corollary 1.7

If $\operatorname{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i}\,\theta(\tau)$.

(Set $z = 0$; for $\tau = it$ this is the functional equation of ϑ.) This gives precise information about θ as $\tau \to 0$; the next corollary describes θ near $\tau = 1$.

Corollary 1.8

If $\tau \in \mathbb{H}$, then

$$\theta(1 - 1/\tau) = \sqrt{\tau/i} \sum_{n=-\infty}^{\infty} e^{\pi i (n+1/2)^2 \tau} = \sqrt{\tau/i}\,(2e^{\pi i \tau/4} + \cdots).$$

The second identity means that $\theta(1-1/\tau) \sim 2\sqrt{\tau/i}\, e^{\pi i \tau/4}$ as $\operatorname{Im}(\tau) \to \infty$.

Proof. n and n^2 have the same parity, so $\theta(1+\tau)=\sum_n(-1)^n e^{\pi i n^2\tau}=\Theta(1/2|\tau)$; hence $\theta(1-1/\tau)=\Theta(1/2|-1/\tau)$. By Theorem 1.6 with $z=1/2$,

$$\theta(1-1/\tau)=\sqrt{\tau/i}\,e^{\pi i\tau/4}\Theta(\tau/2|\tau)=\sqrt{\tau/i}\,e^{\pi i\tau/4}\sum_n e^{\pi i n^2\tau}e^{\pi i n\tau}=\sqrt{\tau/i}\sum_n e^{\pi i(n+1/2)^2\tau}.$$

The terms $n=0,-1$ contribute $2e^{\pi i\tau/4}$, of absolute value $2e^{-\pi t/4}$ ($\tau=\sigma+it$); the others total $O(\sum_{k\ge1}e^{-\pi(k+1/2)^2 t})=O(e^{-9\pi t/4})$. ∎

The **Dedekind eta function** is, for $\mathrm{Im}(\tau)>0$,

$$\eta(\tau)=e^{\frac{\pi i\tau}{12}}\prod_{n=1}^{\infty}(1-e^{2\pi i n\tau}).$$

Proposition 1.9

If $\mathrm{Im}(\tau)>0$, then $\eta(-1/\tau)=\sqrt{\tau/i}\,\eta(\tau)$.

Proof. Differentiate (5) and evaluate at $z_0=1/2+\tau/2$. By the product formula,

$$\Theta(z|\tau)=(1+qe^{-2\pi iz})\prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1}e^{2\pi iz})(1+q^{2n+1}e^{-2\pi iz}),$$

and the first factor vanishes at z_0 (where $e^{2\pi iz_0}=-q$), with derivative $-2\pi iqe^{-2\pi iz_0}=2\pi i$ there, while the remaining factors equal $(1-q^{2n})^3$. So

$$\Theta'(z_0|\tau)=2\pi iH(\tau),\qquad H(\tau)=\prod_{n=1}^{\infty}(1-e^{2\pi i n\tau})^3.$$

Replacing $-1/\tau$ by τ in (5), $\Theta(z|\tau)=\sqrt{i/\tau}\,e^{-\pi iz^2/\tau}\Theta(-z/\tau|-1/\tau)$. Differentiating and evaluating at z_0 (where $-z_0/\tau=-\frac{1}{2\tau}-\frac12$ is a zero of $\Theta(\cdot|-1/\tau)$, congruent to $\frac12+\frac{-1/\tau}{2}$), a computation gives

$$2\pi iH(\tau)=\sqrt{i/\tau}\,e^{-\frac{\pi i\tau}{4}}e^{-\frac{\pi i}{2}}e^{-\frac{\pi i}{4\tau}}\Big(-\frac{1}{\tau}\Big)2\pi iH(-1/\tau),\qquad\text{i.e.}\qquad e^{\frac{\pi i\tau}{4}}H(\tau)=\Big(\frac{i}{\tau}\Big)^{3/2}e^{-\frac{\pi i}{4\tau}}H(-1/\tau),$$

which says $\eta(\tau)^3=(i/\tau)^{3/2}\eta(-1/\tau)^3$. For $\tau=it$, $\eta(\tau)>0$, so taking cube roots gives $\eta(\tau)=\sqrt{i/\tau}\,\eta(-1/\tau)$; by analytic continuation this holds on $\mathbb{H}$. ∎

2 Generating functions

The **generating function** of $\{F_n\}_{n\ge0}$ is $F(x)=\sum_n F_nx^n$. The **partition function** $p(n)$ counts the ways to write n as a sum of positive integers ($p(0)=1$; $p(1)=1$, $p(2)=2$, $p(3)=3$, $p(4)=5$, $p(5)=7$, $p(6)=11$, $p(7)=15$, $p(8)=22$, $p(12)=77$).

Theorem 2.1 (Euler)

If $|x|<1$, then $\displaystyle\sum_{n=0}^{\infty}p(n)x^n=\prod_{k=1}^{\infty}\frac{1}{1-x^k}$.

Formally, expand $\frac{1}{1-x^k}=\sum_m x^{km}$ and multiply: a partition $n=m_1k_1+\cdots+m_rk_r$ (distinct k_j) gives the term $(x^{k_1})^{m_1}\cdots(x^{k_r})^{m_r}$. This is justified as for the product formula of ζ (Chapter 7, Section 1), since $\frac{1}{1-x^k}=1+O(x^k)$.

Similarly $\prod 1/(1-x^{2n-1})$ generates $p_o(n)$, the number of partitions of n into odd parts, and $\prod(1+x^n)$ generates $p_u(n)$, partitions into unequal parts. Remarkably $p_o(n) = p_u(n)$, i.e.

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty}(1+x^n)\,:$$

indeed $\prod(1+x^n)\prod(1-x^n) = \prod(1-x^{2n})$, and by parity $\prod(1-x^{2n})\prod(1-x^{2n-1}) = \prod(1-x^n)$.

Let $p_{e,u}(n)$, $p_{o,u}(n)$ count partitions of n into an even, resp. odd, number of unequal parts. Euler: $p_{e,u}(n) = p_{o,u}(n)$ unless $n = k(3k+1)/2$, $k \in \mathbb{Z}$, is **pentagonal** $(1, 2, 5, 7, 12, 15, 22, 26, \dots)$, when $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$. Multiplying out gives terms $(-1)^r x^{n_1+\cdots+n_r}$ with distinct n_j, so

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{n=0}^{\infty}[p_{e,u}(n) - p_{o,u}(n)]x^n,$$

so Euler's theorem follows from:

Proposition 2.2 (Euler's pentagonal number theorem)

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{k=-\infty}^{\infty}(-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. Set $x = e^{2\pi iu}$. In the triple product take $q = e^{3\pi iu}$ and $z = 1/2 + u/2$; then

$$\prod_{n=1}^{\infty}(1-e^{2\pi i3nu})(1-e^{2\pi i(3n-1)u})(1-e^{2\pi i(3n-2)u}) = \prod_{n=1}^{\infty}(1-e^{2\pi inu}).$$

By Theorem 1.3 the product equals $\sum_n e^{3\pi in^2u}(-1)^n e^{2\pi inu/2} = \sum_n (-1)^n e^{\pi in(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}$. ∎

The growth of $p(n)$ follows from $1/\prod(1-x^n)$ as $|x| \to 1$; the transformation law of η (Proposition 1.9) gives a precise asymptotic formula (Appendix A).

3 The theorems about sums of squares

$r_2(n)$ counts $(x, y) \in \mathbb{Z}^2$ with $n = x^2 + y^2$ (e.g. $r_2(3) = 0$, $r_2(5) = 8$), $r_4(n)$ the representations as sums of four squares. No $4k+3$ is a sum of two squares, no $8k+7$ of three. Which n are sums of two squares, is every n a sum of four, and what are r_2, r_4? Jacobi's theta functions first answered these fully.

3.1 The two-squares theorem

If $n = a^2 + b^2$ and $m = c^2 + d^2$, then with $x + iy = (a + ib)(c + id)$, $x^2 + y^2 = nm$; so divisibility properties of n matter. Let $d_1(n)$ and $d_3(n)$ be the numbers of divisors of n of the form $4k+1$ and $4k+3$.

Theorem 3.1 (Two-squares theorem)

If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.

So $n = p_1^{a_1} \cdots p_r^{a_r}$ is a sum of two squares iff every $p_j \equiv 3 \pmod 4$ has even a_j. For $q = e^{\pi i\tau}$, $\tau \in \mathbb{H}$,

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n)q^n, \tag{6}$$

since $\theta(\tau)^2 = \sum_{n_1}\sum_{n_2} q^{n_1^2}q^{n_2^2} = \sum_{(n_1,n_2)\in\mathbb{Z}^2} q^{n_1^2+n_2^2}$.

Proposition 3.2

The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities

$$\theta(\tau)^2 = 2\sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4\sum_{n=1}^{\infty} \frac{q^n}{1+q^{2n}}, \tag{7}$$

whenever $q = e^{\pi i\tau}$ and $\tau \in \mathbb{H}$.

Proof. Both series converge absolutely ($|q| < 1$), and they are equal since $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$. Since $(1+q^{2n})^{-1} = (1-q^{2n})/(1-q^{4n})$, the right side of (7) is $1 + 4\sum_{n\geq 1}\left[\frac{q^n}{1-q^{4n}} - \frac{q^{3n}}{1-q^{4n}}\right]$. Now $\frac{1}{1-q^{4n}} = \sum_{m\geq 0} q^{4nm}$, so

$$\sum_{n=1}^{\infty}\frac{q^n}{1-q^{4n}} = \sum_{n=1}^{\infty}\sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k)q^k, \qquad \sum_{n=1}^{\infty}\frac{q^{3n}}{1-q^{4n}} = \sum_{k=1}^{\infty} d_3(k)q^k$$

(the series converge since $d_j(k) \leq k$). Compare with (6). ∎

Let

$$C(\tau) = 2\sum_{n=-\infty}^{\infty}\frac{1}{q^n+q^{-n}} = \sum_{n=-\infty}^{\infty}\frac{1}{\cos(\pi n\tau)}, \qquad q = e^{\pi i\tau},\ \tau\in\mathbb{H}. \tag{8}$$

We must prove $\theta^2 = C$. In parallel: θ comes from the heat kernel $e^{-\pi x^2}$ on the line, C from the strip kernel $1/\cosh \pi x$ (Section 1), both their own Fourier transforms (Chapter 3, Section 2, Example 3), and both transformation laws come from Poisson summation. Indeed $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7), and (5) of Chapter 4 with $a = 0$ gives $\sum_n \frac{1}{\cosh \pi nt} = \frac{1}{t}\sum_n \frac{1}{\cosh(\pi n/t)}$, i.e. $C(\tau) = (i/\tau)C(-1/\tau)$ on $\tau = it$, hence on $\mathbb{H}$. Both tend to 1 as $\mathrm{Im}(\tau) \to \infty$. At the cusp $\tau = 1$: $\theta(1-1/\tau)^2 \sim 4(\tau/i)e^{\pi i\tau/2}$ (Corollary 1.8), and (5) of Chapter 4 with $a = 1/2$ gives $\sum_n \frac{(-1)^n}{\cosh(\pi n/t)} = t\sum_n \frac{1}{\cosh(\pi(n+1/2)t)}$, so by analytic continuation

$$C(1-1/\tau) = \frac{\tau}{i}\sum_n \frac{1}{\cos(\pi(n+1/2)\tau)} = 4\frac{\tau}{i}e^{\pi i\tau/2} + O(|\tau|e^{-3\pi t/2}) \quad (t\to\infty),$$

the main terms being $n = 0, -1$.

Proposition 3.3

The function $C(\tau) = \sum 1/\cos(\pi n\tau)$, defined in the upper half-plane, satisfies (i) $C(\tau+2) = C(\tau)$.
(ii) $C(\tau) = (i/\tau)C(-1/\tau)$.
(iii) $C(\tau) \to 1$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $C(1-1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\mathrm{Im}(\tau)\to\infty$. Moreover, $\theta(\tau)^2$ satisfies the same properties.

Theorem 3.4

Suppose f is a holomorphic function in the upper half-plane that satisfies:
(i) $f(\tau+2) = f(\tau)$, (ii) $f(-1/\tau) = f(\tau)$, (iii) $f(\tau)$ is bounded, then f is constant.

Let $\mathcal{F} = \{\tau \in \overline{\mathbb{H}} : |\mathrm{Re}(\tau)| \leq 1,\ |\tau| \geq 1\}$; the points $\tau = \pm 1$ are its **cusps** (equivalent under $\tau \mapsto \tau + 2$).

Lemma 3.5

Every point of $\mathbb{H}$ maps into $\mathcal{F}$ under a composite of $T_2 : \tau \mapsto \tau + 2$, $S : \tau \mapsto -1/\tau$ and their inverses.

$\mathcal{F}$ is the **fundamental domain** of the group G generated by T_2, S (representatives unique except on $\partial\mathcal{F}$). Elements $g \in G$ are matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $g(\tau) = \frac{a\tau+b}{c\tau+d}$, with integer entries and determinant 1, so

$$\operatorname{Im}(g(\tau)) = \frac{\operatorname{Im}(\tau)}{|c\tau + d|^2}. \tag{9}$$

Proof. By (9), with c, d integers, some $g_0 \in G$ maximizes $\operatorname{Im}(g_0(\tau))$; $T_2^{\pm 1}$ preserve imaginary parts, so some g_1 has $|\operatorname{Re}(g_1(\tau))| \le 1$ and maximal Im. If $|g_1(\tau)| < 1$, $\operatorname{Im}(Sg_1(\tau)) = \operatorname{Im}(g_1(\tau))/|g_1(\tau)|^2$ is larger, a contradiction. So $g_1(\tau) \in \mathcal{F}$. ∎

Proof of Theorem 3.4. Suppose f is not constant. $g(z) = f(\tau)$, $z = e^{\pi i \tau}$, is well defined on the punctured disc (period 2) and bounded, so $\lim_{\operatorname{Im}\tau\to\infty} f(\tau)$ exists, with modulus $< \sup_{\mathcal{F}}|f|$ by maximum modulus. At the cusps (by $f(\tau+2) = f(\tau)$, only $\tau = 1$) the same holds for $\lim f(1-1/\tau)$, by interchanging 1 and ∞: let U_n correspond to $\begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$, fixing 1; with $\nu(\tau) = 1/(1-\tau)$, $\nu^{-1}(\tau) = 1 - 1/\tau$, $U_n = \nu^{-1}T_n\nu$, $T_n(\tau) = \tau + n$. So $U_nU_m = U_{n+m}$ and $U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = T_2S$, every U_n lies in G, and $F(\tau) = f(1 - 1/\tau)$ has period 1; $h(z) = F(\tau)$, $z = e^{2\pi i\tau}$, has a removable singularity at 0. So f attains its maximum inside $\mathbb{H}$, contradicting the maximum principle. ∎

Proof of the two-squares theorem. $f = C/\theta^2$ is holomorphic in $\mathbb{H}$ ($\theta \neq 0$, Corollary 1.4), invariant under T_2 and S, and in $\mathcal{F}$ bounded with limit 1 as $\operatorname{Im}(\tau) \to \infty$ or $\tau \to \pm 1$, by (iii) and (iv) of Proposition 3.3. So $f \equiv 1$ by Theorem 3.4: $\theta^2 = C$. ∎

3.2 The four-squares theorem

Let $\sigma_1^*(n)$ be the sum of the divisors of n that are not divisible by 4.

Theorem 3.6 (Four-squares theorem)

Every positive integer is the sum of four squares, and moreover $r_4(n) = 8\sigma_1^*(n)$ for all $n \ge 1$.

As before $\theta(\tau)^4 = \sum_{n\ge 0} r_4(n)q^n$, $q = e^{\pi i\tau}$, but nothing as simple as C is available; we use a variant of the forbidden Eisenstein series $F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}$ (in n, then m; $(n,m) \neq (0,0)$) of Corollary 2.6, defining

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau), \qquad \tau \in \mathbb{H}, \tag{10}$$

equivalently $E_2^*(\tau) = \sum_m \sum_n \left[\frac{1}{(n+m\tau/2)^2} - \frac{1}{(n/2+m\tau)^2}\right]$; the order of summation is critical.

Proposition 3.7

The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity $\theta(\tau)^4 = -\pi^{-2}E_2^*(\tau)$, where $\tau \in \mathbb{H}$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau) = 1 + 8\sum_{k\ge1}\sigma_1^*(k)q^k$. By Corollary 2.6, $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_{k\ge1}\sigma_1(k)e^{2\pi ik\tau}$. And $\sigma_1^*(n) = \sigma_1(n)$ if $4 \nmid n$, $\sigma_1(n) - 4\sigma_1(n/4)$ if $4 \mid n$ (divisors $d = 4\tilde{d}$ of

$n = 4\tilde{n}$ match divisors $\tilde{d}$ of $\tilde{n}$). With (10),

$$E_2^*(\tau) = -\pi^2 - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1^*(k) e^{\pi i k \tau}.$$

■

So Theorem 3.6 reduces to $\theta^4 = -\pi^{-2} E_2^*$, and the key is that E_2^* has the same modular properties as θ^4.

Proposition 3.8

The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:
(i) $E_2^*(\tau + 2) = E_2^*(\tau)$.
(ii) $E_2^*(\tau) = -\tau^{-2} E_2^*(-1/\tau)$.
(iii) $E_2^*(\tau) \to -\pi^2$ as $\operatorname{Im}(\tau) \to \infty$.
(iv) $|E_2^*(1 - 1/\tau)| = O(|\tau^2 e^{\pi i \tau}|)$ as $\operatorname{Im}(\tau) \to \infty$. Moreover $-\pi^2 \theta^4$ has the same properties.

(i) is immediate from the definition. Consider F and its *reverse* $\tilde{F}(\tau) = \sum_n \sum_m \frac{1}{(n+m\tau)^2}$ (summing first in m).

Lemma 3.9

The functions F and $\tilde{F}$ satisfy:
(a) $F(-1/\tau) = \tau^2 \tilde{F}(\tau)$, (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$, (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i \tau$.

Proof. (a) follows from the definitions and $(n+m(-1/\tau))^2 = \tau^{-2}(-m+n\tau)^2$. For (b) use $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$ with $\eta(\tau) = q^{1/12}\prod(1-q^{2n})$, $q = e^{\pi i \tau}$. By Proposition 3.2 of Chapter 5, the logarithmic derivative in τ is

$$\frac{\eta'}{\eta}(\tau) = \frac{\pi i}{12} - 2\pi i \sum_{n=1}^{\infty} \frac{n q^{2n}}{1-q^{2n}}, \qquad \sum_{n=1}^{\infty} \frac{n q^{2n}}{1-q^{2n}} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} n q^{2nm} = \sum_{k=1}^{\infty} \sigma_1(k) q^{2k}.$$

Since $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_k \sigma_1(k) q^{2k}$, $\frac{\eta'}{\eta}(\tau) = \frac{i}{4\pi} F(\tau)$. By the chain rule the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}\frac{\eta'}{\eta}(-1/\tau)$, which by (a) equals $\frac{i}{4\pi}\tilde{F}(\tau)$. Taking logarithmic derivatives in the functional equation for η,

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi} F(\tau),$$

so $\tilde{F}(\tau) = F(\tau) - 2\pi i/\tau$.

(c) follows from (a) and (b). ■

Proof of Proposition 3.8. (ii) By (10) and Lemma 3.9(c),

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) = [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i \tau] \\ &= 4\tau^2 F(2\tau) - \tau^2 F(\tau/2) = -\tau^2 (F(\tau/2) - 4F(2\tau)) = -\tau^2 E_2^*(\tau). \end{aligned}$$

(iii) In $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum \sigma_1(k) e^{2\pi i k \tau}$ the sum tends to 0 as $\operatorname{Im}(\tau) \to \infty$, so by (10) $E_2^*(\tau) \to \frac{\pi^2}{3} - 4\frac{\pi^2}{3} = -\pi^2$.

(iv) We first show

$$E_2^*(1 - 1/\tau) = \tau^2 \Big[F\Big(\frac{\tau - 1}{2}\Big) - F(\tau/2) \Big]. \tag{11}$$

F has period 1, and Lemma 3.9(c) can be written $F(w) = w^{-2}[F(-1/w) + 2\pi i w]$. Let $u = (\tau - 1)/2$ and $w = u/\tau = \frac{1}{2} - \frac{1}{2\tau}$. Then $-1/w = -2 - \frac{2}{\tau - 1}$, so $F(-1/w) = F(-1/u) = u^2 F(u) - 2\pi i u$, and

$$F\Big(\frac{1}{2} - \frac{1}{2\tau}\Big) = \frac{\tau^2}{u^2}\big[u^2 F(u) - 2\pi i u + 2\pi i u/\tau\big] = \tau^2 F(u) - \frac{2\pi i \tau^2 (1 - 1/\tau)}{u} = \tau^2 F\Big(\frac{\tau - 1}{2}\Big) - 4\pi i \tau.$$

Also $4F(2-2/\tau) = 4F(-1/(\tau/2)) = 4[(\tau/2)^2 F(\tau/2) - 2\pi i\tau/2] = \tau^2 F(\tau/2) - 4\pi i\tau$. Subtracting gives (11), since $E_2^*(1-1/\tau) = F(\frac{1}{2} - \frac{1}{2\tau}) - 4F(2-\frac{2}{\tau})$. By the expansion of F, $F(\frac{\tau-1}{2}) - F(\tau/2) = -8\pi^2 \sum_k \sigma_1(k) e^{\pi i k\tau}((-1)^k - 1) = 16\pi^2 e^{\pi i\tau} + O(|e^{3\pi i\tau}|)$, which with (11) gives (iv). For θ^4: (i)–(iii) follow from Section 1 ($\theta(-1/\tau)^4 = (\tau/i)^2\theta(\tau)^4 = -\tau^2\theta(\tau)^4$, $\theta \to 1$), and (iv) from Corollary 1.8: $\theta(1-1/\tau)^4 \sim 16(\tau/i)^2 e^{\pi i\tau}$. ■

Proof of the four-squares theorem. $f = E_2^*/\theta^4$ is holomorphic in $\mathbb{H}$ and invariant under T_2, S (Proposition 3.8). As $\theta(\tau)^4 \to 1$ and $\theta(1-1/\tau)^4 \sim -16\tau^2 e^{\pi i\tau}$, f is bounded in $\mathcal{F}$, hence in $\mathbb{H}$, so constant (Theorem 3.4), equal to $-\pi^2$ by (iii). So $\theta^4 = -\pi^{-2} E_2^*$; in particular $r_4(n) \geq 8 > 0$ for $n \geq 1$. ■